Ex.No.

Date:

Data cleaning and preparation- DataFrame

Aim:

To study and implement different techniques for handling missing values in a Pandas DataFrame using functions such as dropna(), fillna(), replace(), isnull(), notnull(), and forward/backward filling methods, in order to improve data quality and reliability for data analysis and machine learning applications.

Description:

Missing data is always a problem in real life scenarios. Areas like machine learning and data mining face severe issues in the accuracy of their model predictions because of poor quality of data caused by missing values. In these areas, missing value treatment is a major point of focus to make their models more accurate and valid.

Handling Missing Values (Say NA Or Nan) Using Pandas - DataFrame.

1. Drop missing values using dropna() function

2. Fill the missing values using fillna() function

3. Replacing missing values using replace() with a scalar value is equivalent behavior of the fillna() function.

4. Through isnull() and notnull() functions, we can identify the NaN as Boolean result.

5. Fill the missing values from Forward and Backward values through pad/ffill and bfill/backfill functions respectively.

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Data Cleaning and preparation – Excel

Aim:

To perform data cleaning on an Excel dataset using Pandas by handling missing values, removing unwanted spaces, modifying columns and rows, and extracting specific records. Finally, to save the cleaned and updated dataset back into an Excel file.

Description:

Read the Excel file, do the data cleaning process and write the updated data set into excel file

1. Remove the white space using str.strip function
2. Fill the forward values for NaN using fillna(pad) method
3. Drop one particular column using drop function
4. Drop NaN rows using dropna function
5. Replace the values(s) using replace function
6. Extract the particular record based on the isin() function condition

Type your text

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Data Wrangling 1

Aim:

To do Data Wrangling functions.

Description:

Data wrangling is a task in data science and analysis which includes operations like:

Data sorting: To rearrange values in ascending or descending order.
Data filtration: To create a subset of available data.
Data reduction: To eliminate or replace unwanted values.
Data access: To read or write data files.
Data processing: To perform aggregation, statistical and similar operations on specific values.

1. Using join function to join two DataFrame.
2. Using combine function to combine two DataFrame.
3. Using merge function to merge two DataFrame.
4. Using melt function to reshape the DataFrame dimension.

Data Wrangling 2

Ex.No.

Date:

Aim:

To do Data Wrangling functions.

Description:

1. Using replace function to replace the NaN values by average value.
2. Filtering and dropping the rows and columns respectively.
3. Using concat function to concatenate two DataFrame.

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Aim:

To do Data Wrangling functions.

Description:

1. Using groupby function to group the data set.
2. Using duplicated function to remove duplicated rows in the DataFrame.
3. Using merge function to merge two DataFrame.

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Data Aggregation and Grouping

Aim:

To perform Data Aggregation and grouping functions

Note:

Aggregation: Aggregation means applying statistical operations like **sum,mean,min,max,count** on data. It is used to **summarize the data** by performing calculations. It uses the function agg(). **Grouping:** Grouping means **splitting the data into groups** based on a column or condition. It is used **organize data into categories** before performing operations. It uses the function groupby().

Description:

1. Create a DataFrame.
2. Implement Data Aggregation and Grouping functions

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Data Aggregation

Aim:

To perform Data Aggregation functions

Description:

1. Create a DataFrame.
2. Implement the following Data Aggregation functions.
   -Aggregate or agg function
   -columnwise and row wise aggregate function
   -sum,min,max,mean
   -Describe

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Data Grouping

Aim:

To perform Data Grouping functions
Description:

1. Create a DataFrame.
2. Implement the following Data Grouping functions.
   -Mean
   -Sum
   -First
   -Last
   -Count
   -Size

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Time Series 1

Aim:

To implement and check Time series functions in python

Description:

1. Get today's, previous day and next day dates
2. Display the data format from the raw data
3. Replacing the day/month/year with replace function
4. Display day/month/year from current date

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Time Series 2

Aim:

To implement and check Time series functions in python

Description:

1. Different date strings formats are checked

Ex.No.

Date:

Time Series 3

Aim:

To implement and check Time series functions in python

Description:

1. Implement different date time function through DataFrame.

Ex.No.

Date:

Time Series 4

Aim:

To implement and check Time series functions in python

Description:

1. Date range function with periods and frequency

Ex.No.

Date:

Time Series 5

Aim:

To implement and check Time series functions in python

Description:

1. Through datetime index, perform truncate, duplicates and repeated date colums

Ex.No.

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Data Visualization 1

Aim:

To import Data Visualization packages and do the basic functions.

Description:

1. Get the X-axis and Y-axis points through Numpy array and plot the line.
2. Plot the points without line using the following ‘marker’ options.
   Marker options:
   ‘O’ – circle
   ‘*’ – Star
   ‘.’ – point
   ‘d’ – Diamond (thin)
   ‘D’ – Diamond
   ‘s’ – Square
   ‘p’ – plus
   ‘+’ – plus
   ‘H’ or ‘h’ – Hexagon
   ‘p’ – pentagon
   ‘v’ – Triangle down
   ‘ʌ’ – Triangle up
   ‘<’ – Triangle left
   ‘>’ – Triangle right
3. Multiple points options also can be plotted
4. Using default X-axis value by specifying only Y-axis value, we can plot points.

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Data Visualization 2

Aim:

To implement different matplot functions.

Description:

1. Use title, X-axis and Y-axis labels
   Labels:
   xlabel()
   ylabel()
   title()
      fontdict=font1
      loc = ‘left’ or ‘right’ or ‘center’
2. Use line style, colors and line width as below.
   Line style
   ‘-’ – solid line
   ‘:’ – Dotted line (or) linestyle = ‘dotted’ ‘.’ – point
   ‘__’ – Dashed line (or) linestyle = ‘dashed’
   ‘-.’ – Dashed, dotted line
   Color style
   ‘r’ – Red
   ‘g’ – Green
   ‘b’ – Blue
   ‘c’ – Cyan
   ‘m’ – Magenta
   ‘y’ – Yellow
   ‘k’ – Black
   ‘w’ – White
   Example for the above options
   Plt.plot(ypoints,’o:r’)
   Plt.plot(ypoints,marker=’o’,ms=20,mec=’r’)
   ms – marker size
   mec – marker edge color
   mfc – marker face color

3. Use multiple lines with grid options
4. Use sub-plots options
5. Use scatter options
   -Single coordinates
   -Double coordinates

Ex.No.

Data Visualization 3

Date:

Aim:

To do other matplot options.

Description:

1. Draw bar chart
2. Draw multi-bar chart with title x and y labels
3. Draw horizontal bar
4. Use color and width options
5. Draw histogram
6. Draw pie chart
   -without label
   -with label
   -with explode

Ex.No.

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Data Visualization 4

Aim:

To implement other Data Visualization tools/packages

Description:

1. Using seaborn package to do lineplot, scatterplot and barplot
2. Draw 3-D plot
3. Using plotly package to do scatterplot